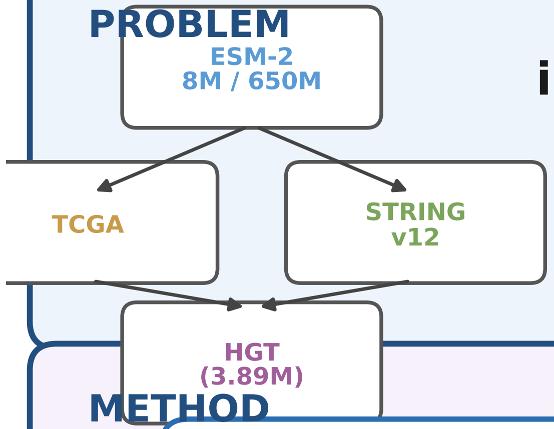
PROBLEM



Which graph SSL objective drives encoder quality in the realistic PLM-initialized multi-cancer regime?

Motivation · Literature gap

PLM features (ESM-2) on graph nodes — task ranking unknown
No protocol to diagnose task-loss □ encoder-quality decoupling

METHOD

Equal-budget 5-seed ablation · 3 tasks · HGT encoder

3 TASKS

o NAMR	Denoising reconstruction (feature scaling mandatory)
o CSP	PPI link prediction (-> collapse diagnostic)
o C3L	Cross-cancer InfoNCE (-> redundancy elimination)

5 x EQUAL-BUDGET

> **Seeds 42 / 7 / 123 / 21 / 99**	n = 5 same-code runs
> **500 epochs per arm**	equal optimizer steps
> **Equal-volume 4,998-edge subgraph**	FLOP parity vs BRCA-only

LEVEL-1/2/3

❶ **Level-1 PPI AUC**	SOLE ranking metric
❷ **Level-2 cluster accuracy**	Self-consistency only
❸ **Level-3 downstream**	Scope boundaries only

FINDINGS

3 Guidelines + Scope Boundaries + 10-command Reproduction

GUIDELINE 1 [STRONG] NAMR reconstruction wins (when features are scaled)

GUIDELINE 2 [STRONG] Encoder collapse <> CSP loss (skip-decoder diagnostic)

GUIDELINE 3 [CONDITIONAL] ESM-2 => C3L redundancy (feature-first design rule)

HONEST SCOPE BOUNDARIES (3 / 3)

Gene-level relational tasks GAIN
Patient-level phenotype tasks (set-fingerprint) NOISE
Drug response from target-only embeddings R^2 < 0

REPRODUCIBILITY : 10-command recipe

R/RunMultiCancerSSL.R 5 verbs
Makefile (data/preprocess/train/evaluate/figures)
96-row checkpoint provenance registry JSON